

MENTORA ACADEMY

DEPARTMENT OF ADVANCED CHEMISTRY · CAMBRIDGE INTERNATIONAL A LEVEL (9701)

Inorganic Chemistry Experimental Suite

Planning, Analysis and Evaluation · Transition Metal Ligand Stoichiometry · Group 2 Decomposition Kinetics

CANDIDATE NAME:	Urwah	CENTRE NUMBER:	PK082	CANDIDATE NO:	0142
COMPONENT:	Paper 5 – Planning, Analysis and Evaluation	DURATION:	1 Hour 15 Minutes	MAX MARKS:	30 Marks

READ THESE INSTRUCTIONS FIRST

- Write your name, Centre number, and candidate number in the boxes above.
- Answer all questions in the spaces provided on the question paper.
- In Question 1 (Planning): define independent, dependent, and controlled variables; describe step-by-step procedures with apparatus diagrams; calculate required quantities; state hazard risk assessments.
- In Question 2 (Analysis, Conclusions & Evaluation): process raw experimental data into derived columns; quote values to consistent significant figures; plot graphs with appropriate linear scales; determine gradients and intercepts; identify anomalous points; evaluate uncertainties.
- You may use an approved scientific calculator. The Periodic Table is provided as standard reference.

EXAMINATION BREAKDOWN & SYLLABUS SKILLS TESTED

Question	Investigation Title & Focus	Syllabus Ref	Type of Task	Marks
Question 1	Planning a Colorimetric Determination of the Formula of an Iron(III)–Thiocyanate Complex	9701/52/M/J/23/ Q1 · Syllabus 28.2	PLANNING	15
Question 2	Analysis and Evaluation of the Thermal Decomposition of Basic Copper(II) Carbonate	9701/51/O/N/23/Q 2 · Syllabus 27.1 / 28.2	ANALYSIS & EVALUATION	15

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QUESTION 1

PLANNING

[Total: 15 Marks]

Planning a Colorimetric Determination of the Formula of an Iron(III)–Thiocyanate Complex

Cambridge Syllabus Reference: 9701/52/M/J/23/Q1 · Syllabus 28.2

When aqueous iron(III) ions react with aqueous thiocyanate ions, SCN, an intensely blood-red complex is formed:

$$\text{Fe}^3(\text{aq}) + n \text{SCN}(\text{aq}) \rightarrow [\text{Fe}(\text{SCN})_n]^3(\text{aq})$$

The formula and coordination stoichiometry, n , can be determined using Job's method of continuous variation. In this method, a series of mixtures of equimolar solutions of $\text{Fe}^3(\text{aq})$ and $\text{SCN}(\text{aq})$ are prepared such that the total volume ($V_{\text{Fe}} + V_{\text{SCN}}$) is held strictly constant, while the mole fraction of SCN varies from 0.0 to 1.0.

The absorbance of each mixture is measured using a colorimeter equipped with a green/blue complementary filter (approx. 480 nm). The absorbance reaches a maximum when the reactants are mixed in their exact stoichiometric ratio.

You are provided with: $0.0020 \text{ mol dm}^{-3} \text{ Fe}(\text{NO})$ in $0.1 \text{ mol dm}^{-3} \text{ HNO}_3$, $0.0020 \text{ mol dm}^{-3} \text{ KSCN}(\text{aq})$, distilled water, a digital colorimeter with matched cuvettes, burettes, graduated pipettes, and standard glassware.

- (a) State the independent variable and the dependent variable in this continuous variation investigation. [2]

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- (b) Describe how you would prepare a series of nine reaction mixtures of total volume 25.0 cm^3 having mole fractions of SCN ranging from 0.10 to 0.90. [3]

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- (c) Explain why a green/blue filter (~480 nm) is selected for the colorimeter, and describe how the colorimeter is calibrated before taking measurements. [3]

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- (d) Explain how the student determines the value of n from a graph of absorbance against mole fraction of SCN. [3]

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- (e) Identify two potential sources of experimental error in this colorimetric method and suggest practical precautions to mitigate them. [4]

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QUESTION 2

ANALYSIS & EVALUATION

[Total: 15 Marks]

Analysis and Evaluation of the Thermal Decomposition of Basic Copper(II) Carbonate

Cambridge Syllabus Reference: 9701/51/O/N/23/Q2 · Syllabus 27.1 / 28.2

Basic copper(II) carbonate, malachite, has the formula $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$. On strong heating, it decomposes completely according to the stoichiometric equation:



A student carried out gravimetric thermal decomposition experiments on eight different masses of pure malachite in porcelain crucibles using a Bunsen burner.

Data collected: Initial mass of malachite sample, m , and final mass of black copper(II) oxide residue, m , after heating to constant mass.

Theoretical molar masses: Malachite = 221.1 g mol⁻¹; CuO = 79.5 g mol⁻¹; 2 moles of CuO produced per mole of malachite = 159.0 g.

- (a) Calculate the theoretical percentage mass of CuO residue remaining when pure malachite decomposes completely. [2]

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- (b) A graph of mass of CuO residue (m / g) on the y-axis against initial mass of malachite (m / g) on the x-axis was plotted. [3]
- (i) State the expected theoretical gradient of the straight line.
- (ii) State whether the line should pass through the origin, explaining why.

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- (c) In one experiment, a student recorded: $m = 3.50 \text{ g}$ of malachite and obtained $m = 2.85 \text{ g}$ of residue. [4]
- Explain why this point lies significantly ABOVE the line of best fit, identifying two possible experimental errors.

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- (d) Describe the precise experimental procedure of 'heating to constant mass' to ensure complete decomposition. [3]

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- (e) Suggest how the gaseous products could be trapped and quantified to provide an independent cross-check on the decomposition stoichiometry. [3]

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Comprehensive Examiner Mark Scheme

Marking points, indicative responses, and error-carried-forward (ECF) guidelines

MARK SCHEME: QUESTION 1 (PLANNING) — Planning a Colorimetric Determination of the Formula of an Iron(III)–Thiocyanate Complex

[15 Marks]

Part	Examiner Marking Points / Indicative Content	Marks
(a)	Independent variable: Mole fraction / volume ratio of SCN (or volume of KSCN added) [1]; Dependent variable: Absorbance of the solution measured on the colorimeter [1].	[2]
(b)	Use two clean, calibrated burettes: one filled with 0.0020 mol dm ³ Fe(NO) and one with 0.0020 mol dm ³ KSCN [1]; Measure volumes such that V _{Fe} + V _{SCN} = 25.0 cm ³ for each mixture (e.g. 2.5 cm ³ KSCN + 22.5 cm ³ Fe ³ for x=0.10, 5.0 + 20.0 for x=0.20, up to 22.5 + 2.5 for x=0.90) [1]; Mix thoroughly in separate beakers and allow to stand for 2 minutes to establish chemical equilibrium [1].	[3]
(c)	The complex is blood-red; green/blue light (480 nm) is the complementary color that is strongly absorbed by the complex, providing maximum sensitivity [1]; Fill a clean cuvette with distilled water (or Fe ³ blank solution) [1]; Place in colorimeter and press calibration button to set absorbance to 0.00 (100% transmission) [1].	[3]
(d)	Plot absorbance on y-axis against mole fraction of SCN (x _{SCN}) on x-axis [1]; Draw two straight lines of best fit through the rising and falling data points and locate their point of intersection [1]; Read the value of x _{max} at the intersection: stoichiometry ratio n = x _{max} / (1 - x _{max}) (e.g. if x _{max} = 0.50, n = 1 forming [Fe(SCN)] ²⁺) [1].	[3]
(e)	Error 1: Fingerprints / scratches on optical faces of the cuvette altering light transmission [1]; Precaution: Wipe optical faces with lint-free lens tissue and handle cuvettes only by ribbed/frosted sides [1]; Error 2: Hydrolysis of Fe ³ (aq) forming colored colloidal iron(III) hydroxides [1]; Precaution: Maintain acidic conditions (0.1 mol dm ³ HNO ₃) in all Fe ³ solutions to suppress hydrolysis [1].	[4]

MARK SCHEME: QUESTION 2 (ANALYSIS & EVALUATION) — Analysis and Evaluation of the Thermal Decomposition of Basic Copper(II) Carbonate

[15 Marks]

Part	Examiner Marking Points / Indicative Content	Marks
(a)	Theoretical % mass = $[2 \times \text{Mr}(\text{CuO}) / \text{Mr}(\text{malachite})] \times 100\% = [159.0 / 221.1] \times 100\% = 71.9\%$ [2].	[2]
(b)	(i) Theoretical gradient = $159.0 / 221.1 = 0.719$ [1]; (ii) Yes, the line must pass through the origin (0,0) [1]; zero mass of malachite will produce zero mass of copper(II) oxide residue [1].	[3]
(c)	Observed % residue = $(2.85 / 3.50) \times 100\% = 81.4\%$, which is much higher than theoretical 71.9% [1]; Error 1: Incomplete decomposition (sample was not heated for sufficient time or to a high enough temperature to expel all CO ₂ and H ₂ O) [1]; Error 2: Failure to heat to constant mass / crucible was weighed hot creating convection currents / crucible absorbed atmospheric moisture before weighing [2].	[4]
(d)	Heat the crucible and contents strongly for 5 minutes, allow to cool in a desiccator, and record the mass [1]; Reheat strongly for another 2–3 minutes, cool in the desiccator, and re-weigh [1]; Repeat the cycle until two consecutive mass readings agree to within ± 0.002 g [1].	[3]
(e)	Pass the evolved gases through a U-tube containing anhydrous calcium chloride to absorb and weigh H ₂ O(g) [1]; Subsequently bubble the gas through weighed aqueous sodium hydroxide / soda lime U-tube to absorb and weigh CO ₂ (g) [1]; Verify that mass loss of solid equals sum of masses of absorbed H ₂ O and CO ₂ [1].	[3]